Ericsson PE Node BAS-PE-10 - Standard Operating Procedure for Resolving Hardware/Environmental Faults (ASIC Memory Parity Errors, Line Card Crashes, Power Supply Redundancy Loss) | ALARM_5160 - Minor Severity

Node: BAS-PE-10 | **Vendor:** Ericsson | **Severity:** Minor | **Alarm Code:** ALARM_5160

Description

This Standard Operating Procedure (SOP) addresses the resolution of hardware/environmental faults, specifically ASIC memory parity errors, line card crashes, and power supply redundancy loss, on the Ericsson PE Node BAS-PE-10 at the Basavanagudi POP. The underlying root cause of this failure domain is attributed to the inherent design of the PE Node's hardware architecture, which utilizes a distributed architecture for traffic processing. In this implementation, the PE Node's line cards are responsible for handling traffic forwarding, and the ASICs (Application-Specific Integrated Circuits) embedded within these line cards perform memory-intensive operations. As a result, the likelihood of memory parity errors occurring due to hardware malfunctions or environmental stressors is increased. Furthermore, the power supply redundancy mechanism, designed to ensure continued operation in case of a single power supply failure, may also be compromised in the event of a hardware fault, leading to a loss of redundancy and exacerbating the impact of the failure. This SOP aims to provide a structured approach to identifying and resolving such failures, thereby minimizing downtime and ensuring the continued operation of the PE Node.

Pre-requisite Checks

- Verify that the remote hands team has been dispatched to the Basavanagudi POP to provide physical access to the PE Node.

- Check the status of the redundant link to ensure that it is operational and can be utilized as a fallback in case of a failure.

- Obtain the current alarm log and review it for any relevant information related to the ALARM_5160 error.

Potential Root Causes

Cause	Probability
ASIC memory parity error due to hardware malfunction	High
Line card crash due to software bug or hardware failure	Medium
Power supply redundancy loss due to hardware failure	Low

Diagnostic Commands & Expected Outputs

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Command: show alarm-alarm, Expected Output: ALARM_5160 - ASIC memory parity error, Description: This command retrieves the current alarm log and displays the ALARM_5160 error, which indicates an ASIC memory parity error. The expected output will provide information on the time of occurrence, the node ID, and the alarm severity.

Command: show card-status all, Expected Output: Line card X is in error state, Description: This command displays the status of all line cards on the PE Node. The expected output will indicate which line card is in error state, providing valuable information for further troubleshooting.

Command: show power-supply status, Expected Output: Redundancy loss detected, Description: This command displays the status of the power supplies on the PE Node. The expected output will indicate if redundancy loss has been detected, which may be the root cause of the failure.
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Resolution Steps

Step 1: Step: Replace faulty line card, Command: replace card X, Description: This command initiates the replacement of the faulty line card with a spare unit. The PE Node will automatically recognize the new line card and reconfigure its resources.

Step 2: Step: Restore power supply redundancy, Command: power-supply redundancy restore, Description: This command restores the power supply redundancy mechanism, ensuring that the PE Node can continue to operate in case of a single power supply failure.

Post-Check Verification

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Command: show card-status all, Expected Output: All line cards are in normal state, Description: This command verifies that all line cards on the PE Node are in normal state, indicating that the replacement was successful.

Command: show power-supply status, Expected Output: Redundancy restored, Description: This command verifies that the power supply redundancy mechanism has been restored, ensuring continued operation in case of a single power supply failure.
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Roll-back Procedures

- Step: Revert line card replacement, Command: replace card X revert, Description: This command reverts the replacement of the faulty line card, restoring the original faulty unit.

- Step: Revert power supply redundancy restoration, Command: power-supply redundancy restore revert, Description: This command reverts the restoration of the power supply redundancy mechanism, restoring the original state.